DELIVERY DELAY & LOGISTICS ANALYSIS

Data-Driven Insights for Delivery Reliability & Customer Trust

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1. Project Overview

This report presents a comprehensive analysis of delivery delays and logistics performance for an e-commerce platform. The analysis leverages transactional order data — spanning orders, sellers, products, payments, and geolocation — to identify the root causes of late deliveries, unreliable delivery estimates, and their impact on customer trust.

The platform's current estimated delivery dates are frequently over-conservative — often twice the actual delivery time — causing potential customers to abandon purchases before checkout. Simultaneously, when carriers genuinely run late, customers lose trust and do not return. This dual problem is what we refer to as the "Trust Gap."

1.1 Business Problem Statement

> *"What is the actual versus estimated delivery time, and how does this delay impact customer review scores? How can we use machine learning to provide a more accurate — and shorter — delivery estimate to increase conversion without over-promising?"*

1.2 Dataset Summary

The analysis is based on six primary MySQL tables ingested from CSV files into the delivery_performance database:

- orders — Order lifecycle timestamps, statuses, and estimated delivery dates
- order_items — Line-level product, seller, price, and freight data
- payments — Payment type, installments, and payment values per order
- products — Product category, dimensions, and weight
- sellers — Seller city, state, and zip code
- geolocation — Geographic coordinates mapped to zip codes

2. Data Ingestion & Database Setup

Raw CSV files were loaded into a MySQL database using Python. Key libraries used: Pandas, SQLAlchemy, OS, Logging, and Time. A dedicated ingestion script automated the loading process and maintained detailed execution logs.

2.1 Ingestion Script

```
# Import Required Libraries
import pandas as pd
from sqlalchemy import create_engine
import os, logging, time

# MySQL Connection
engine = create_engine(
    f'mysql+pymysql://root:***@localhost:3306/delivery_performance'
)

def ingestion_db(df, tablename, engine):
    df.to_sql(tablename, con=engine, if_exists='replace', index=False)

def load_row_data():
    start = time.time()
```

```
    for file in os.listdir('data_files'):
        if '.csv' in file:
            df = pd.read_csv('data_files/' + file)
            logging.info(f'Ingesting {file} into database')
            ingestion_db(df, file[:-4], engine)
    logging.info(f'Total Time: {(time.time()-start)/60:.2f} min')
```

3. Exploratory Data Analysis & Data Preparation

This section covers data exploration, table inspection, null-value handling, feature engineering, and the creation of the consolidated delivery_delay_summary analytical table.

3.1 Table Exploration

A loop was used to verify all loaded tables and their record counts:

```
tables = pd.read_sql_query(
    'SHOW TABLES FROM delivery_performance', con=engine
)

for table in tables['Tables_in_delivery_performance']:
    count = pd.read_sql_query(
        f'SELECT COUNT(*) AS Count FROM {table}', con=engine
    )['Count'].values[0]
    print(f'{table}: {count} records')
    display(pd.read_sql(f'SELECT * FROM {table} LIMIT 5', con=engine))
```

Key record counts identified:

Table	Total Records	Unique Order IDs
order_items	112,650	98,666
orders	99,441	99,441
payments	103,886	99,440
sellers	3,095	3,095
products	32,951	32,951

3.2 Consolidated Delivery Summary Table

A unified analytical table — delivery_delay_summary — was built using a multi-CTE SQL query joining all five source tables. This single query-ready structure powers all downstream analysis.

```
-- Main CTE: Join order_items, payments, products, sellers, orders
WITH customer_order_info AS (
    SELECT
        ord.order_id, ord.seller_id, ord.shipping_limit_date,
        ord.price, ord.freight_value,
        py.payment_type, py.payment_installments, py.payment_value,
        pd.product_category, pd.product_weight_g,
        se.seller_city, se.seller_state,
        ords.order_status, ords.order_purchase_timestamp,
        ords.order_approved_at, ords.order_delivered_carrier_date,
        ords.order_delivered_customer_date,
        ords.order_estimated_delivery_date
    FROM order_items ord
    JOIN payments py ON ord.order_id = py.order_id
    JOIN products pd ON ord.product_id = pd.product_id
```

```
    JOIN sellers se ON ord.seller_id = se.seller_id
    JOIN orders ords ON ord.order_id = ords.order_id
),
location_info AS (
    SELECT IFNULL(se.seller_id,'NA') AS seller_id,
           ge.geolocation_city, ge.geolocation_state
    FROM geolocation ge
    LEFT JOIN sellers se
    ON ge.geolocation_zip_code_prefix = se.seller_zip_code_prefix
)
```

3.3 Feature Engineering

Several derived metrics were computed to enable multi-dimensional delay analysis:

- order_handling_days — Days from order_purchase_timestamp to order_approved_at
- order_approved_days — Days from approval to carrier handoff
- order_delivered_customer_days — Total days from purchase to customer delivery
- order_delivered_late — Boolean flag: 1 if delivered after estimated date
- is_late_shipping — Boolean flag: 1 if delivered after shipping_limit_date

```
-- Feature Engineering in SQL
SELECT *,
    DATEDIFF(order_approved_at, order_purchase_timestamp)
        AS order_handling_days,
    DATEDIFF(order_delivered_carrier_date, order_approved_at)
        AS order_approved_days,
    DATEDIFF(order_delivered_customer_date, order_purchase_timestamp)
        AS order_delivered_customer_days,
    CASE WHEN order_delivered_customer_date
              > order_estimated_delivery_date THEN 1 ELSE 0 END
        AS order_delivered_late,
    CASE WHEN order_delivered_customer_date
              > shipping_limit_date THEN 1 ELSE 0 END
        AS is_late_shipping
FROM customer_order_info
```

4. Data Analysis — Python & SQL

4.1 Late Deliveries by Product Category

SQL was used to count the total number of orders delivered late per product category, revealing which categories suffer most from delivery failures.

```
SELECT product_category,
       COUNT(*) AS late_order_delivered
FROM delivery_delay_summary
WHERE order_delivered_late = 1
GROUP BY product_category
ORDER BY late_order_delivered DESC
LIMIT 5;
```

SQL Output:

product_category	late_order_delivered
bed table bath	739
HEALTH BEAUTY	675
sport leisure	509
Furniture Decoration	470
computer accessories	429

Figure 1: Top 5 Product Categories by Late Delivery Count

Bed Table Bath leads with 739 late deliveries, followed by Health Beauty (675) and Sport Leisure (509). These high-volume, potentially bulky or fragile product categories appear most susceptible to carrier and handling delays.

4.2 Sellers with Most Late Deliveries

Sellers responsible for the highest frequency of late order deliveries were identified, enabling targeted seller performance interventions.

```
SELECT seller_id,
       COUNT(*) AS times_late
FROM delivery_delay_summary
WHERE order_delivered_late = 1
GROUP BY seller_id
ORDER BY times_late DESC
LIMIT 10;
```

SQL Output:

seller_id	times_late
7c67e1448b00f6e969d365cea6b010ab	200
88460e8ebdecbfecb5f9601833981930	118
1025f0e2d44d7041d6cf58b6550e0bfa	117
06a2c3af7b3aee5d69171b0e14f0ee87	105
8160255418d5aaa7dbdc9f4c64ebda44	99
3d871de0142ce09b7081e2b9d1733cb1	68
4a3ca9315b744ce9f8e9374361493884	67
1f50f920176fa81dab994f9023523100	60
37515688008a7a40ac93e3b2e4ab203f	57
7aa4334be125fcdd2ba64b3180029f14	57

Figure 2: Top 10 Sellers by Frequency of Late Deliveries

The top seller shows 200 late deliveries — nearly double the next highest (118). These sellers represent high-priority candidates for seller performance reviews, SLA enforcement, or re-routing to faster carriers.

4.3 Sellers with Most Late Shipments

Beyond delivery date, we also tracked sellers whose shipments exceeded the shipping_limit_date — a different failure point focused on carrier handoff delays.

```
SELECT seller_id,
       COUNT(*) AS late_shipped_delivered
FROM delivery_delay_summary
WHERE is_late_shipping = 1
GROUP BY seller_id
ORDER BY late_shipped_delivered DESC
LIMIT 5;
```

SQL Output:

seller_id	late_shipped_delivered
06a2c3af7b3aee5d69171b0e14f0ee87	68
1025f0e2d44d7041d6cf58b6550e0bfa	41
88460e8ebdecbfecb5f9601833981930	40
8160255418d5aaa7dbdc9f4c64ebda44	34
7c67e1448b00f6e969d365cea6b010ab	32

Figure 3: Top 5 Sellers by Late Shipment Count

Seller 06a2c3af shows 68 late shipments — suggesting a persistent carrier handoff issue specific to this seller's logistics pipeline.

4.4 Late Shipments by Product Category

```
SELECT product_category,
       COUNT(*) AS late_shipped_delivered
FROM delivery_delay_summary
WHERE is_late_shipping = 1
GROUP BY product_category
ORDER BY late_shipped_delivered DESC
LIMIT 5;
```

SQL Output:

product_category	late_shipped_delivered
HEALTH BEAUTY	173
bed table bath	149
computer accessories	148
Furniture Decoration	147
housewares	89

Figure 4: Top 5 Product Categories by Late Shipment Count

Health Beauty (173) and Bed Table Bath (149) top the late shipment rankings as well, reinforcing that these categories face systemic logistics challenges end-to-end — from seller handoff to final customer delivery.

4.5 Average Order Approval Days by Payment Type

We examined whether payment method influences how quickly an order gets approved and enters the fulfillment pipeline.

```
SELECT payment_type,
       ROUND(AVG(order_handling_days), 4) AS avg_days
FROM delivery_delay_summary
GROUP BY payment_type
ORDER BY avg_days DESC;
```

SQL Output:

payment_type	avg_days
credit_card	0.1943
UPI	1.7749
voucher	0.3219
debit_card	0.3783

Figure 5: Average Order Handling Days by Payment Type

UPI payments show significantly higher average handling time (1.77 days) compared to credit card (0.19 days), vouchers (0.32 days), and debit cards (0.38 days). This suggests UPI payment confirmation latency is a contributing bottleneck in the order approval pipeline.

4.6 Delivery Days by Freight Value Tier

Orders were segmented by freight value into four tiers to understand the relationship between shipping cost, product weight, and delivery speed.

```
SELECT
    ROUND(AVG(order_delivered_customer_days), 4) AS avg_order_delivery_days,
    ROUND(AVG(product_weight_g), 4) AS avg_product_weight,
    CASE
        WHEN total_freight_value BETWEEN 0 AND 10
            THEN 'low_freight_value(0-10)'
        WHEN total_freight_value BETWEEN 10 AND 20
            THEN 'standard_freight_value(10-20)'
        WHEN total_freight_value BETWEEN 20 AND 30
            THEN 'high_freight_value(20-30)'
        WHEN total_freight_value BETWEEN 30 AND 40
            THEN 'very_high_freight_value(30-40)'
```

```
    END AS freight_value
FROM delivery_delay_summary
GROUP BY freight_value
ORDER BY avg_order_delivery_days;
```

SQL Output:

avg_order_delivery_days	avg_product_weight	freight_value
5.7040	581.021746031746	low_freight_value(0-10)
11.6985	1217.1473118084855	standard_freight_value(10-20)
14.1786	2591.391515008272	high_freight_value(20-30)
15.1836	5682.0110847189235	very_high_freight_value(30-40)

Figure 6: Delivery Days vs Freight Value Tier and Product Weight

A clear positive correlation exists: heavier products with higher freight charges take significantly longer to deliver. Low freight orders (avg weight ~581g) deliver in ~5.7 days, while very high freight orders (avg weight ~5,682g) take ~15.2 days. This finding is critical for building a more accurate ML-based delivery estimator.

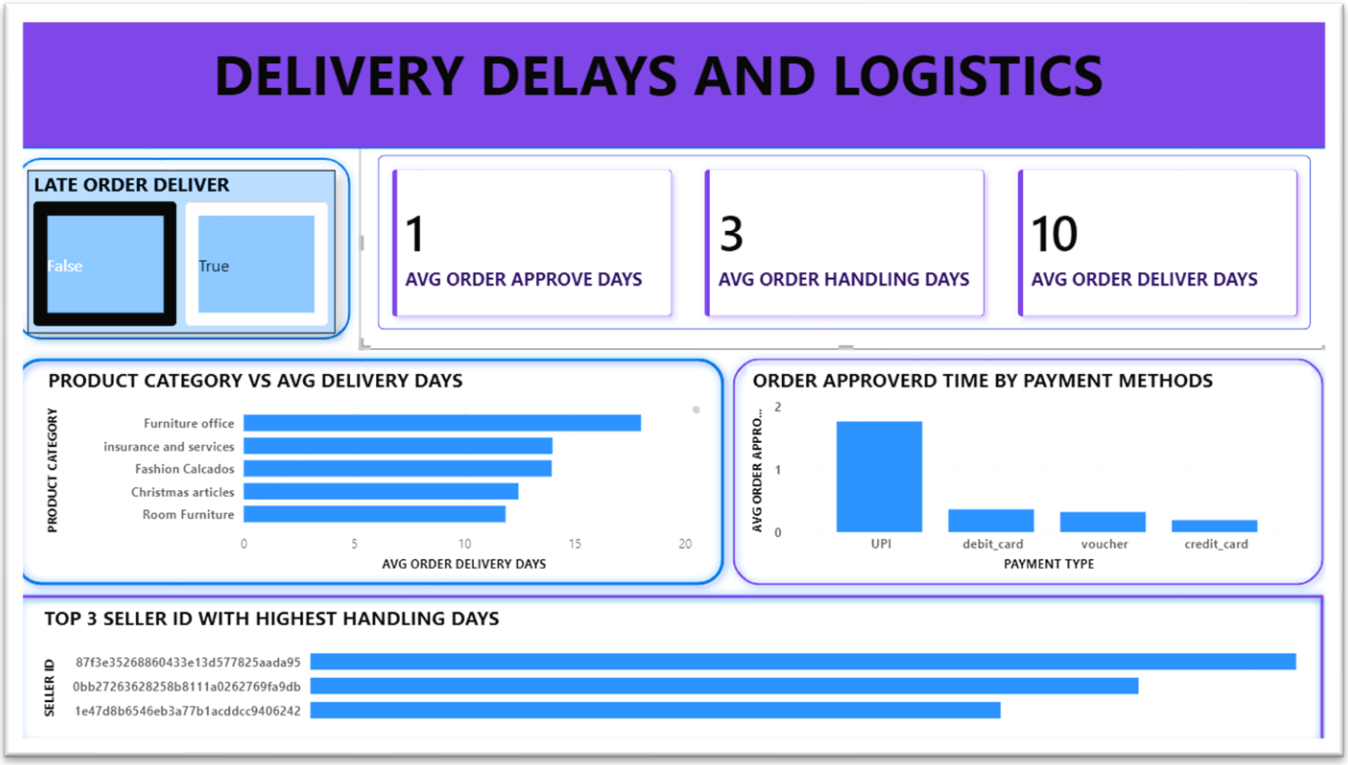
5. Key Findings & Insights

The following summary table consolidates all major findings from the analysis:

Finding Area	Key Insight
Top Late Category	Bed Table Bath (739 late orders) is the worst-performing product category for on-time delivery
Top Late Seller	Seller 7c67e1448b had 200 late deliveries — nearly 2x the second-highest seller
Payment Bottleneck	UPI payments average 1.77 days to approval vs 0.19 days for credit card — a 9x gap
Weight & Freight	Heavier products (5,682g avg) take ~15.2 days to deliver vs 5.7 days for lightweight items
The Trust Gap	Estimated delivery dates are frequently 2x the actual delivery time, reducing conversion rates
Late Shipping Risk	Health Beauty and Bed Table Bath dominate both late delivery AND late shipping categories

6. Data Visualization Power Bi

An interactive Power BI dashboard was developed to consolidate analytical findings into a visually compelling and user-friendly reporting interface. The dashboard enables stakeholders to dynamically explore product delivery delay metrics, filter by category, top seller ID, payment types and monitor key KPIs in real time.



7. Business Recommendations

Based on the analysis conducted, the following strategic recommendations are proposed:

1. **Penalize & Coach Chronically Late Sellers**

The top 10 sellers by late delivery frequency should be placed on performance improvement plans. SLA enforcement with financial penalties for persistent lateness, combined with logistics coaching, can significantly reduce the late delivery rate.

2. **Address UPI Payment Approval Latency**

The 9x gap in order handling time between UPI (1.77 days) and credit card (0.19 days) suggests an integration or banking API bottleneck. Optimizing the UPI confirmation pipeline or pre-clearing orders can reduce this delay and improve overall fulfillment speed.

3. **Build a Weight-Aware Delivery Estimator**

Product weight and freight value are the strongest predictors of delivery time. An ML model incorporating these features, alongside seller location and category, can produce estimates far more accurate than the current conservative defaults — increasing customer conversion without overpromising.

4. **Category-Specific Logistics Partnerships**

Bed Table Bath and Health Beauty are consistently problematic across both late delivery and late shipping metrics. Dedicated carrier partnerships or fulfillment center prioritization for these categories should be evaluated.

5. **Implement Real-Time Delivery Monitoring Dashboard**

Operationalize the findings through a BI dashboard that tracks at-risk orders in real time — flagging orders where delivery is trending late before the customer delivery date is missed. This enables proactive intervention rather than retrospective damage control.

The Trust Gap is a Leaky Bucket: *Every late delivery costs not just one order — it costs future orders from the same customer. Fixing delivery reliability is the highest-ROI investment an e-commerce platform can make in long-term customer retention.*